

Mechanics of Solid (M.O.S)
#3130608



Dedicated Faculty, Committed Education

Darshan
Institute of Engineering & Technology

Unit-3

Moment of Inertia & Mass Moment of Inertia

Mechanics of Solids
(3130608)



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Syllabus – Module 3 Description



GUJARAT TECHNOLOGICAL UNIVERSITY

Bachelor of Engineering

Subject Code: 3130608

Semester III

Mechanics of Solids

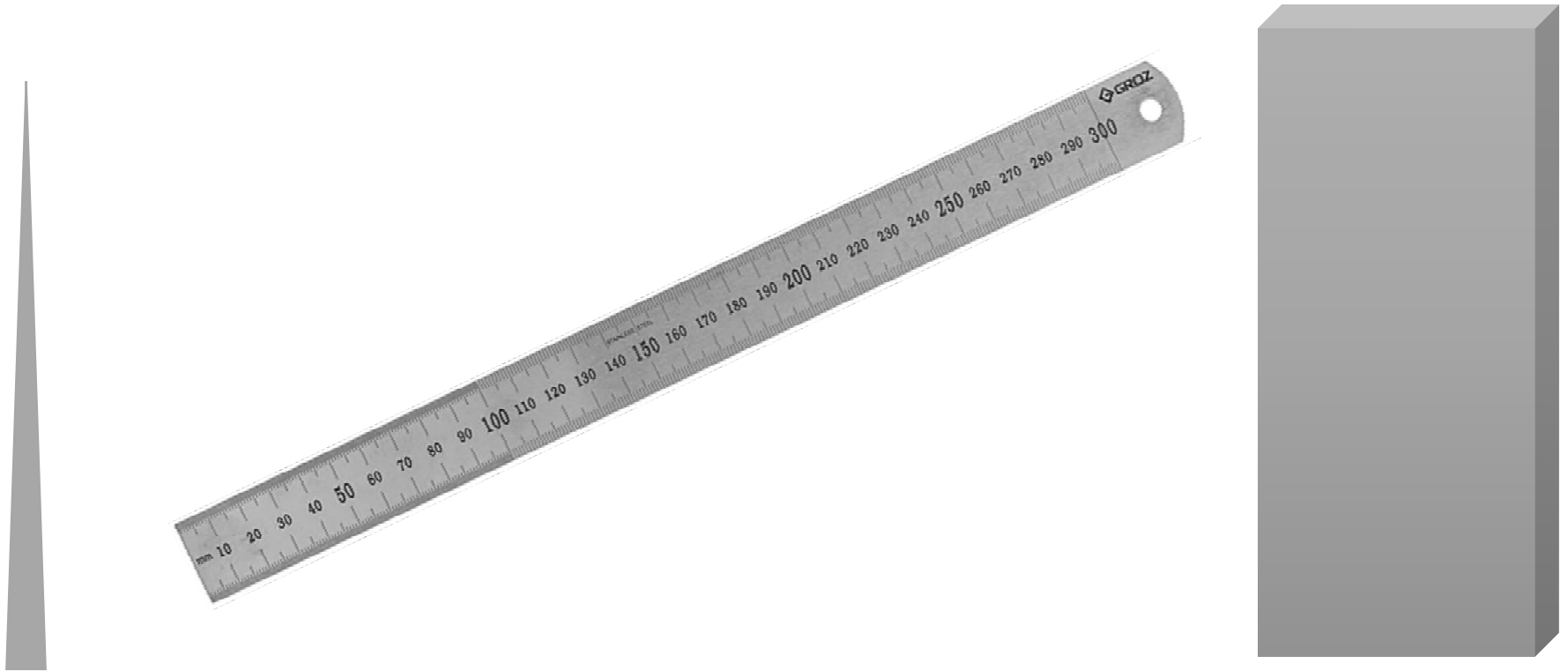
MODULE 3

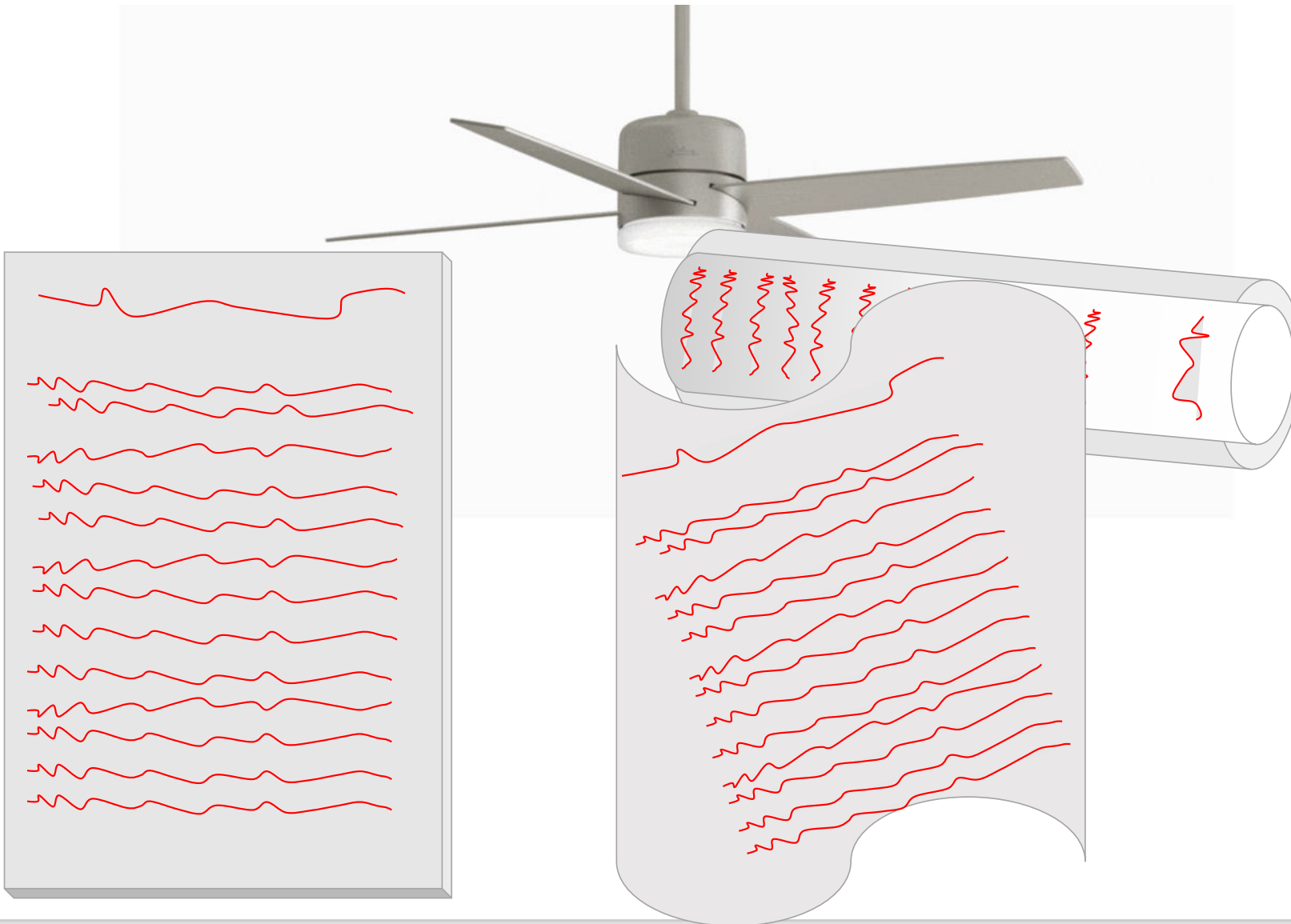
5	Centroid and moment of inertia and mass moment of inertia Centroid: Centroid of lines, plane areas and volumes, Examples related to centroid of composite geometry, Pappus – Guldinus first and second theorems. Moment of inertia of planar cross-sections: Derivation of equation of moment of inertia of standard lamina using first principle, Parallel & perpendicular axes theorems, polar moment of inertia, radius of gyration of areas, section modulus. Examples related to moment of inertia of composite geometry	08	15
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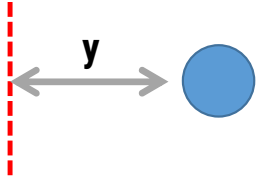
Understanding of Moment of Inertia







Introduction



Moment of area = Area $\times$ perpendicular distance

$$\longrightarrow M = A \times y$$

- ▶ If the moment of area is taken about same reference axis, it is known as moment of inertia in terms of area, which is defined as,

Moment of inertia = moment $\times$ perpendicular distance

$$I_A = M \times y = A \times y \times y = Ay^2$$

- ▶ Where I_A is area moment of inertia, A is area and ' y ' is the distance between centroid of area and reference axis. On similar notes, moment of inertia is also determined in terms of mass, which is defined as,
- ▶ Where ' m ' is mass of body, ' r ' is distance between center of mass of body and reference axis and I_m is mass moment of inertia about reference axis. It must be noted here that for same area or mass moment of inertia will change with change in location of reference axis.

$$I_m = mr^2$$

Theorem of Parallel Axis

- It states, If the moment of inertia of a plane area about an axis through its center of gravity is denoted by I_G , then moment of inertia of the area about any other axis AB parallel to the first and at a distance 'h' from the center of gravity is given by,

$$I_{AB} = I_G + ah^2$$

Where,

I_{AB} = Moment of inertia of the area about AB axis

I_G = Moment of inertia of the area about centroid

a = Area of section

h = Distance between center of gravity (centroid) of the section & axis AB

Theorem of Parallel Axis : - Proof

- ▶ Consider a strip of a circle, whose moment of inertia is required to be found out a line 'AB' as shown in figure.

Let's y = Distance of the strip from the C.G. of the section

h = Distance between center of gravity of the section and the 'AB' axis

d_a = Area of the strip

- ▶ We know that moment of inertia of the whole section about an axis passing through the center of gravity of the section.

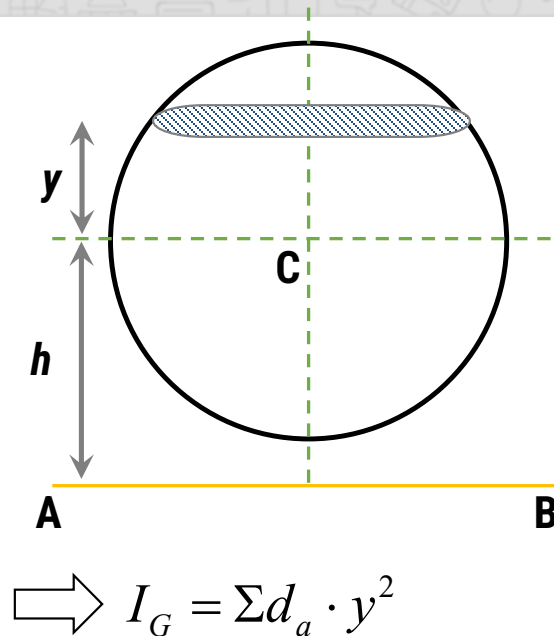
$$I_G = d_a \cdot y^2$$

- ▶ And M.I of the whole section about an axis passing through centroid

- ▶ Moment of inertia of the section about the AB axis

$$I_{AB} = \sum d_a (h + y)^2 = \sum d_a (h^2 + 2 \cdot h \cdot y + y^2) = ah^2 + I_G$$

Note: - $\sum d_a \cdot h^2 = a \cdot h$, $\sum y^2 d_a = I_G$ & $\sum d_a \cdot y$ is the algebraic sum of moments of all the areas, about an axis through center of gravity of the section and is equal $a\bar{y}$, where $\bar{y}$ is the distance between the section and the axis passing through the center of gravity which obviously is zero



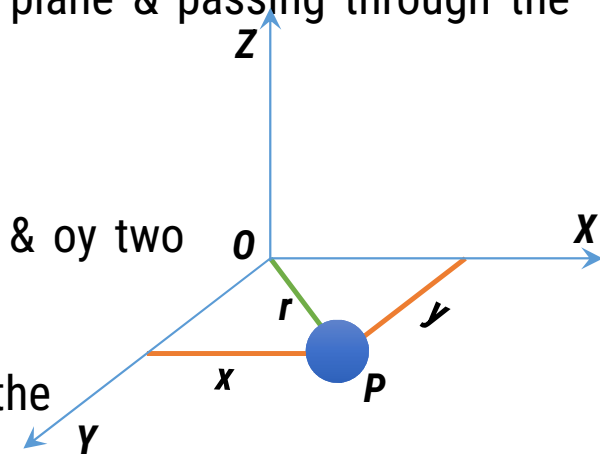
Theorem of Perpendicular Axis

- It states, If I_{xx} and I_{yy} be the moment of inertia of a plane section about two perpendicular axis meeting at 'O' the moment of inertia I_{zz} about the axis z-z, perpendicular to the plane & passing through the intersection of x-x and y-y is given by,

$$I_{zz} = I_{xx} + I_{yy}$$

Proof :- Consider a small lamina (P) of area 'da' having co-ordinates as ox & oy two mutually perpendicular axes on a plane section as shown in fig

Now, consider a plane OZ perpendicular ox and oy. Let (r) be the distance of the lamina (p) from z-z axis such that op = r.



From the geometry of the figure, we find that, $\Rightarrow r^2 = x^2 + y^2$

We know that the moment of inertia of the lamina 'p' about x-x axis

$$I_{xx} = d_a \cdot y^2$$

$$I_{yy} = d_a \cdot x^2$$

$$I_{zz} = d_a \cdot r^2 = d_a (x^2 + y^2) = d_a \cdot x^2 + d_a \cdot y^2 = I_{xx} + I_{yy}$$

Moment of Inertia of a Rectangular Section

- ▶ Consider a rectangular section ABCD as shown in fig. whose moment of inertia is required to be found out

Let, b = width of the section, d = Depth of the section

- ▶ Now, consider a strip PQ of thickness dy parallel to x-x axis and at a distance y from it as shown in fig.

$$\text{Area of strip} = b \times dy$$

- ▶ We know that moment of inertia of the strip about x-x axis,

$$= \text{area} \times y^2$$

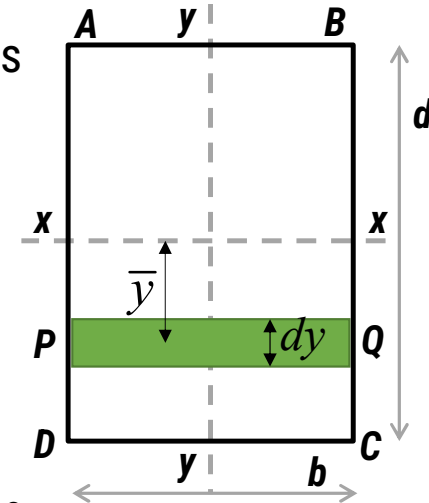
$$= (b \cdot dy) \times y^2$$

- ▶ Now, moment of inertia of the whole section may be found out by integrating the about equation for the whole length of the lamina i.e. from $-d/2$ to $+d/2$

$$I_{xx} = \int_{-d/2}^{+d/2} b \times y^2 dy = b \int_{-d/2}^{+d/2} y^2 dy = b \times \left[\frac{y^3}{3} \right]_{-d/2}^{+d/2} = \frac{b \times d^3}{12}$$

Similarly, $I_{yy} = \frac{d \times b^3}{12}$

- ▶ If it is square section, $I_{xx} = I_{yy} = \frac{d^4}{12} = \frac{b^4}{12}$



Moment of Inertia of a Triangular Section

- Consider a triangular section ABC whose moment of inertia is required to be found out

Let, b = width of the section, h = height of triangle

- Now, consider a small strip PQ of thickness ' dx ' at a distance from the vertex A as shown in figure, we find that the two triangle APQ and ABC are similar.

$$\frac{PQ}{BC} = \frac{x}{h} \Rightarrow PQ = \frac{BC \times x}{h} = \frac{b \times x}{h}$$

- We know that area of the strip $PQ = \frac{b \times x}{h} dx$

- And moment of inertia of the strip about the base BC $= \text{area} \times (\text{distance})^2 = \frac{b \times x}{h} dx (h - x)^2$

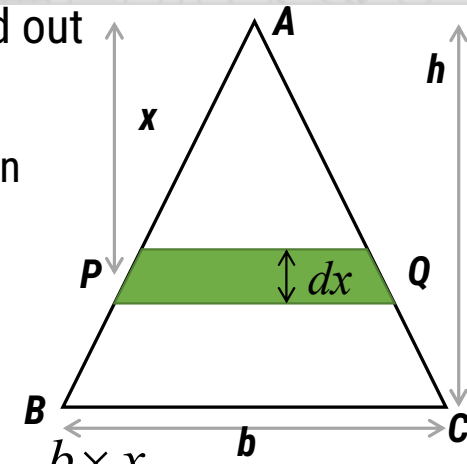
- Now, moment of inertia of the whole triangular section may be found out by integrating the above equation for the above equation for the whole height of the triangle i.e. from 0 to h .

$$I_{BC} = \int_0^h \frac{b \times x}{h} (h - x)^2 dx = \frac{b}{h} \int_0^h (h^2 - 2hx + x^2) x \cdot dx = \frac{b}{h} \left[\frac{h^2 x^2}{2} - \frac{2hx^3}{3} + \frac{x^4}{4} \right]_0^h = \frac{bh^3}{12}$$

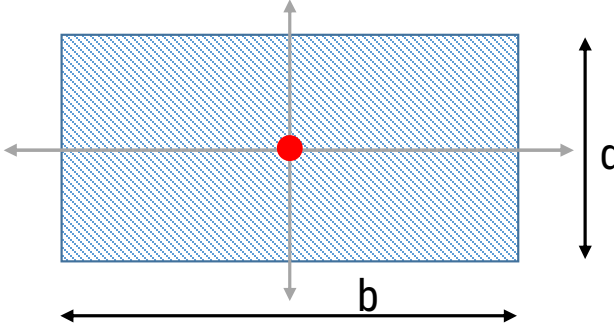
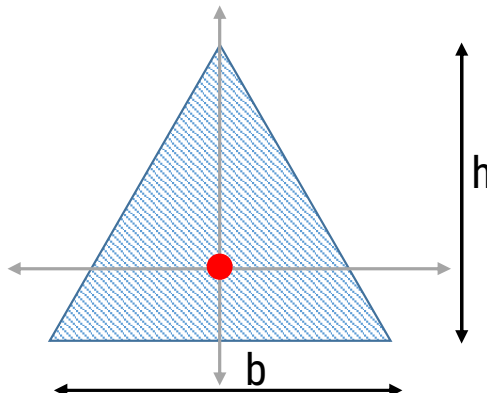
- We know that the distance between center of gravity of the triangular section & Base BC, $y = (h/3)$

- so, Moment of the inertia of the triangular section about an axis through its center through its center of gravity parallel to x-x axis,

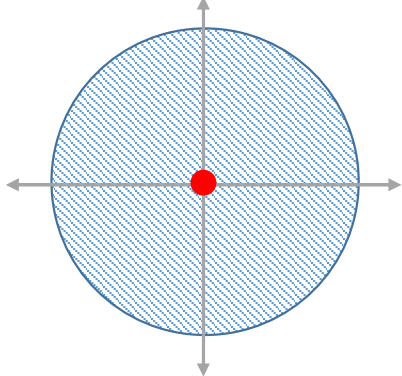
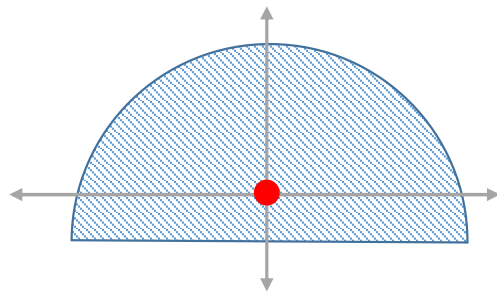
$$I_{xx} = I_{BC} - ay^2 = \frac{bh^3}{12} - \left(\frac{bh}{2} \right) \left(\frac{h}{3} \right)^2 = \frac{bh^3}{36}$$



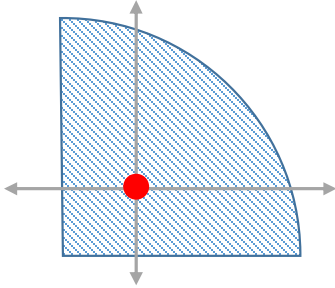
Area (Lamina) Element – Moment of Inertia (Basic Shape)

Element Name	Geometrical Shape	Area	I_{xx}	I_{yy}
Rectangle		$b \times d$	$\frac{bd^3}{12}$	$\frac{db^3}{12}$
Triangle		$\frac{1}{2} \times b \times h$	$\frac{bh^3}{36}$	$\frac{hb^3}{36}$

Area (Lamina) Element – Moment of Inertia (Basic Shape)

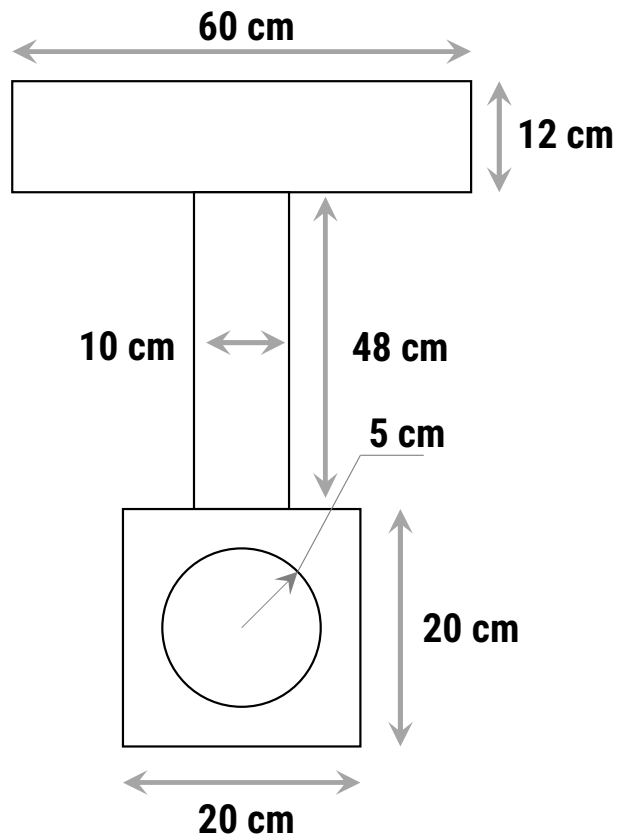
Element Name	Geometrical Shape	Area	I_{xx}	I_{yy}
Circle		πr^2	$\frac{\pi d^4}{64}$	$\frac{\pi d^4}{64}$
Semi circle		$\frac{\pi r^2}{2}$	$0.11r^4$	$\frac{\pi d^4}{128}$

Area (Lamina) Element – Moment of Inertia (Basic Shape)

Element Name	Geometrical Shape	Area	I_{xx}	I_{yy}
Quarter circle		$\frac{\pi r^2}{4}$	$0.055r^4$	$0.055r^4$

Example of Moment of Inertia

Ex - 1 Determine moment of inertia of a section shown in fig about horizontal centroid axis

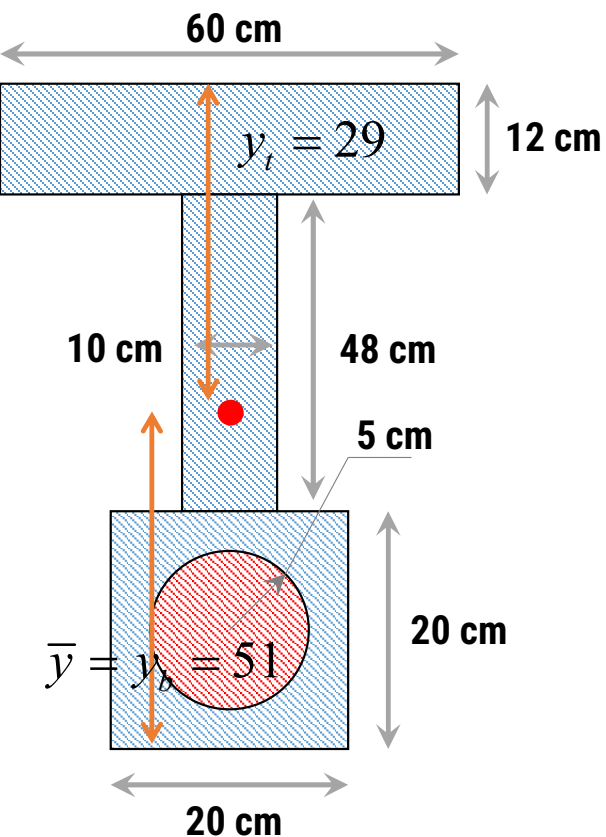


Answer

► **Centroid of given lamina**

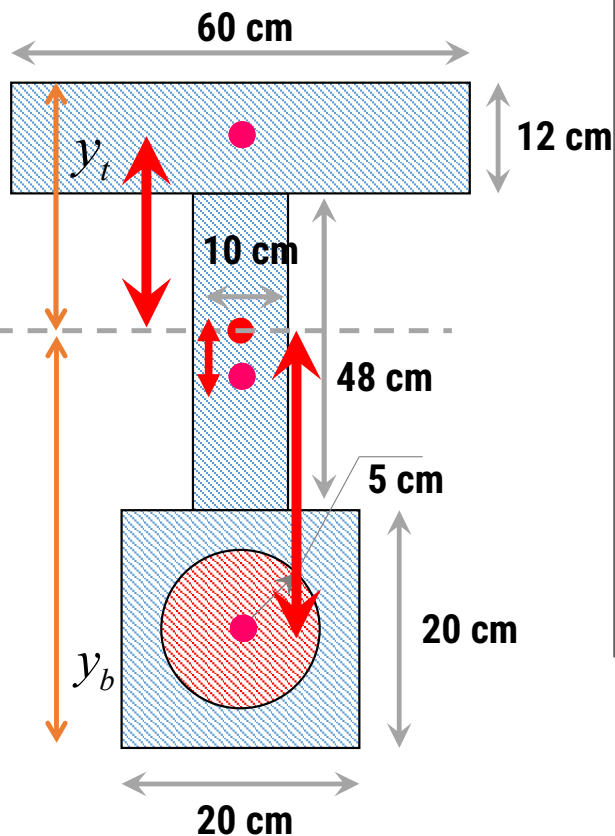
Let's divide the given lamina into four parts

- 1) Top rectangular $60 \times 12 \text{ cm}^2$
- 2) Middle rectangular $10 \times 48 \text{ cm}^2$
- 3) Bottom square $20 \times 20 \text{ cm}^2$
- 4) Deduct circle of radius 5 cm from bottom square



Shape	Area (cm ²)	Y (cm)	AY (cm ³)
Top rectangular	$60 \times 12 = 720$	$20 + 48 + \frac{12}{2} = 74$	53280
Middle rectangular	$48 \times 10 = 480$	$20 + \frac{48}{2} = 44$	21120
Bottom square	$20 \times 20 = 400$	$\frac{20}{2} = 10$	4000
Deduct circle	$-\pi(5^2) = -78.54$	$\frac{20}{2} = 10$	-785.4
$\Sigma A = 1521.46$		$\Sigma A \cdot y = 77614.6$	

$$\bar{y} = \frac{A_1 y_1 + A_2 y_2 + A_3 y_3 - A_4 y_4}{A_1 + A_2 + A_3 - A_4} = \frac{\Sigma A \cdot y}{\Sigma A} = \frac{77614.6}{1521.46} = 51 \text{ cm}$$



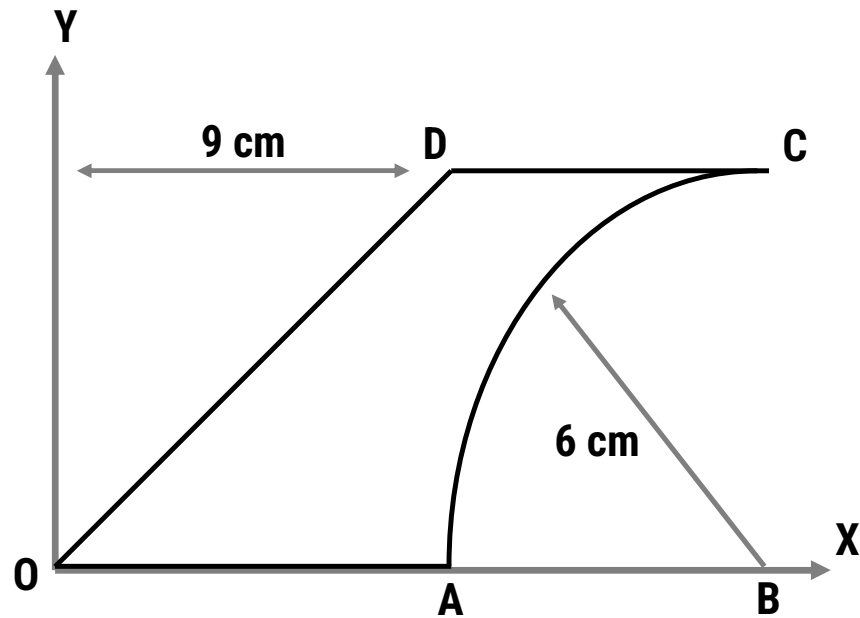
$$y_t = 29 \quad y_b = 51$$

Area	h	Ah ²	I _G	I _{xx} = I _G + Ah ²
720	$29 - \frac{12}{2} = 23$	380.88×10^3	$\frac{60 \times 12^3}{12} = 8640$	389.52×10^3
480	$51 - (12+24) = 15$	180×10^3	$\frac{10 \times 48^3}{12} = 92160$	272.16×10^3
400	$51 - 10 = 41$	672.4×10^3	$\frac{20^4}{12} = 13333.34$	685.73×10^3
-78.54	$51 - 10 = 41$	-132.025×10^3	$-\frac{\pi \times 10^4}{64} = -490.8$	-132.516×10^3

$$I_{xx} = I_1 + I_2 + I_3 - I_4 = 1.214894 \times 10^6 \text{ cm}^2$$

Example of Moment of Inertia

Ex - 2 Find the moment of inertia about Y-axis & X-axis for the area shown in fig.



Answer

► **Centroid of given lamina**

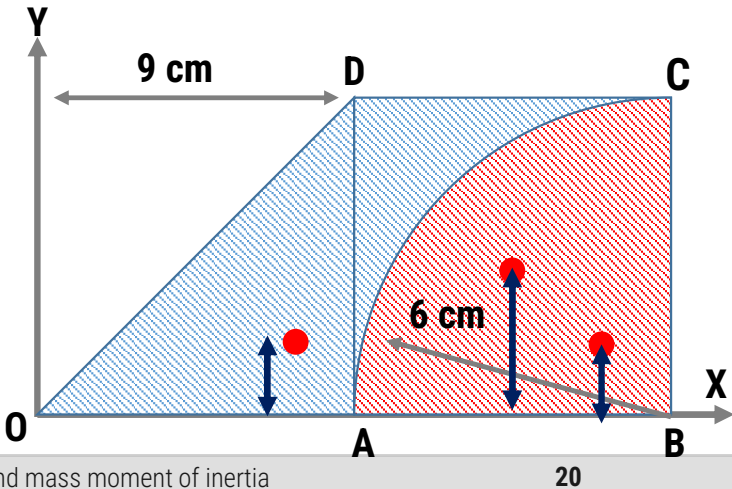
Let's divide the given lamina in to three part

- 1) Triangle (9 cm x 6 cm)
- 2) Square (6 cm x 6 cm)
- 3) Deduct Quarter circle having radius 6 cm

► **Moment of inertia about x- axis (o-x line)**

Shape	Area (cm ²)	h (cm)	Ah ² (cm ⁴)	I _G (cm ⁴)	I _{ox} = I _G + Ah ²
Triangle	$\frac{9 \times 6}{2} = 27$	$\frac{6}{3} = 2$	108	$\frac{9 \times 6^3}{36} = 54$	162
Square	$6 \times 6 = 36$	$\frac{6}{2} = 3$	324	$\frac{6^4}{12} = 108$	432
Quarter circle	$-\frac{\pi \times 6^2}{4} = -28.27$	$\frac{4 \times 6}{3\pi} = 2.55$	183.82	$0.055 \times 6^4 = 71.28$	-255.1

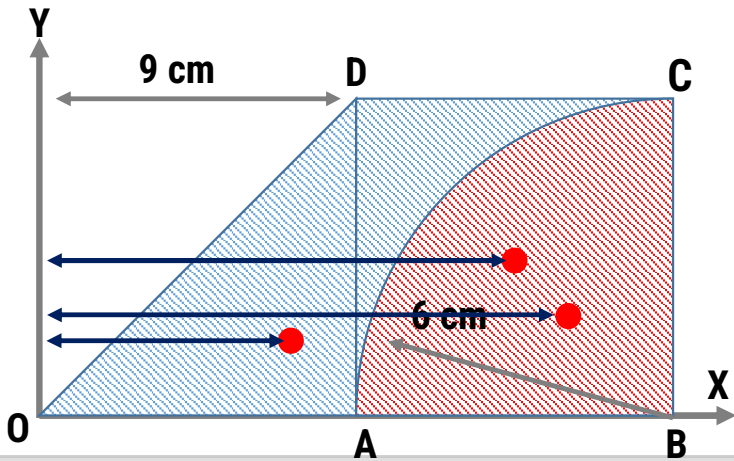
$$I_{ox} = 162 + 431 - 255.1 = 337.9cm^4$$



► **Moment of inertia about y- axis (o-y line)**

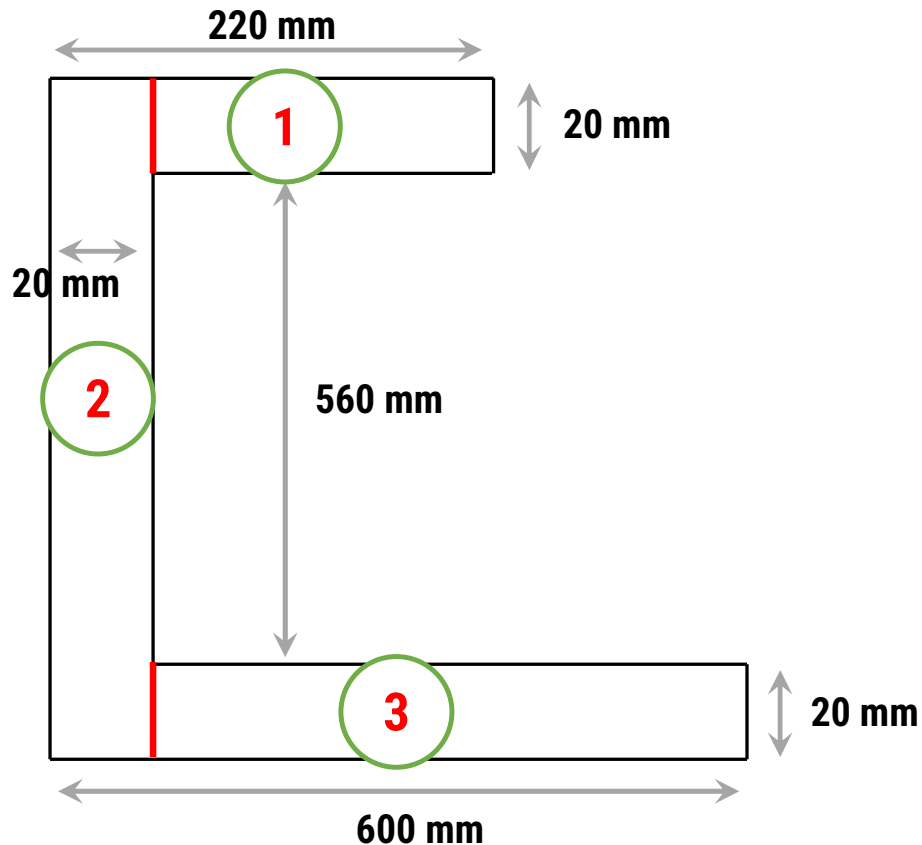
Shape	Area (cm ²)	h (cm)	Ah ² (cm ⁴)	I _G (cm ⁴)	I _{oy} = I _G + Ah ²
Triangle	$\frac{9 \times 6}{2} = 27$	$\frac{2 \times 9}{3} = 6$	972	$\frac{9^3 \times 6}{36} = 121.5$	1093.5
Square	$6 \times 6 = 36$	$\frac{6}{2} + 9 = 12$	5184	$\frac{6^4}{12} = 108$	5292
Quarter circle	$-\frac{\pi \times 6^2}{4} = -28.27$	$9 + 6 - \frac{4 \times 6}{3\pi} = 12.45$	4381.92	$0.055 \times 6^4 = 71.28$	-4453.2

$$I_{oy} = 1093.5 + 5292 - 4453.2 = 1932.3cm^4$$



Example of Moment of Inertia

Ex - 3 Find out moment of inertia at horizontal and vertical centroid axes



Answer

► Centroid of given lamina

Let's divide the given lamina in to three part

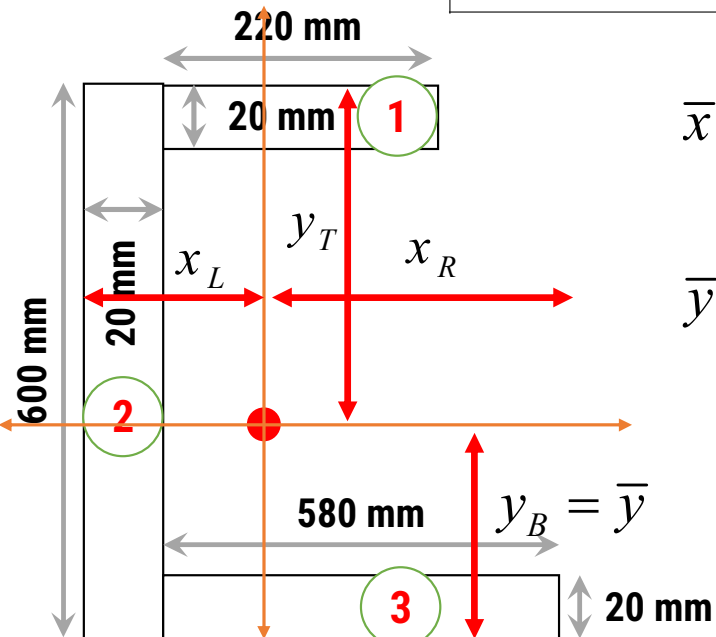
- 1) Top rectangle $200 \times 20 \text{ mm}^2$
- 2) Middle rectangle $20 \times 600 \text{ mm}^2$
- 3) Bottom rectangle $580 \times 20 \text{ mm}^2$

Top rectangle **1**
200 x 20=(4000)mm²

Middle rectangle **2**
600 x 20=(12000)mm²

Bottom rectangle **3**
580 x 20=(11600)mm²

Area (mm ²)	X (mm)	Y (mm)	AX (mm ³)	AY (mm ³)
4000	$20 + \frac{220}{2} = 130$	$20 + 560 + \frac{20}{2} = 590$	520×10^3	2.36×10^6
12000	$\frac{20}{2} = 10$	$\frac{600}{2} = 300$	120×10^3	3.6×10^6
11600	$20 + \frac{580}{2} = 310$	$\frac{20}{2} = 10$	3.596×10^6	116×10^3
$\Sigma A = 27600$		$\Sigma AX = 4.236 \times 10^6$		$\Sigma AY = 6.076 \times 10^6$



$$\bar{x} = \frac{A_1 x_1 + A_2 x_2 + A_3 x_3}{A_1 + A_2 + A_3} = \frac{\Sigma A \cdot x}{\Sigma A} = \frac{4.236 \times 10^6}{27600} = 153.48 \text{ mm}$$

$$\bar{y} = \frac{A_1 y_1 + A_2 y_2 + A_3 y_3}{A_1 + A_2 + A_3} = \frac{\Sigma A \cdot y}{\Sigma A} = \frac{6.076 \times 10^6}{27600} = 220.15 \text{ mm}$$

$$\bar{x} = x_L = 153.48 \quad x_R = 446.52$$

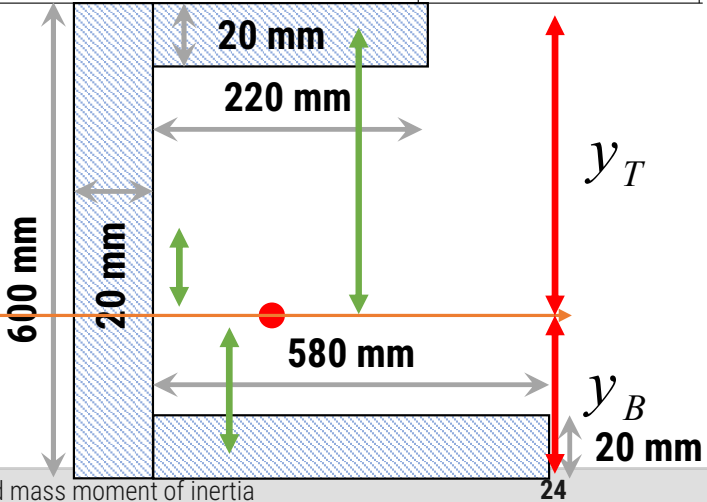
$$\bar{y} = y_B = 220.15 \quad y_T = 379.85$$

► **Moment of inertia about centroid horizontal axis**

Shape	Area (mm ²)	h (mm)	Ah ² (mm ⁴)	I _G (cm ⁴)	I _{xx} = I _G + Ah ²
Top Rectangle	220 × 20 = 4400	$379.85 - \frac{20}{2} = 369.85$	601.87×10^6	$\frac{220 \times 20^3}{12} = 146.67 \times 10^3$	602.017×10^6
Middle Rectangle	600 × 20 = 12000	$379.85 - \frac{600}{2} = 79.85$	7.651×10^6	$\frac{20 \times 600^3}{12} = 360 \times 10^6$	367.651×10^6
Bottom Rectangle	580 × 20 = 11600	$220.15 - \frac{20}{2} = 210.15$	512.29×10^6	$\frac{580 \times 20^3}{12} = 386.67 \times 10^3$	512.68×10^6

$$I_{xx} = I_1 + I_2 + I_3 = 1.4825 \times 10^9 \text{ mm}^4$$

$$\begin{aligned} \bar{y} &= y_B = 220.15 \\ y_T &= 379.85 \end{aligned}$$



► **Moment of inertia about centroid vertical axis**

Shape	Area (mm ²)	h (mm)	Ah ² (mm ⁴)	I _G (mm ⁴)	I _{yy} = I _G + Ah ²
Top Rectangle	220 × 20 = 4400	153.48 – 130 = 23.48	2.426 × 10 ⁶	$\frac{20 \times 220^3}{12} = 17.7467 \times 10^6$	20.1727 × 10 ⁶
Middle Rectangle	600 × 20 = 12000	153.48 – $\frac{20}{2}$ = 143.48	24.704 × 10 ⁶	$\frac{600 \times 20^3}{12} = 400 \times 10^3$	25.104 × 10 ⁶
Bottom Rectangle	580 × 20 = 11600	446.52 – $\frac{580}{2}$ = 156.52	284.2 × 10 ⁶	$\frac{20 \times 580^3}{12} = 325.187 \times 10^6$	609.387 × 10 ⁶

$$I_{yy} = I_1 + I_2 + I_3 = 654.6637 \times 10^6 \text{ mm}^4$$

$$x_L = 153.48$$

$$x_R = 446.52$$

